

Selenium + TestNG Cheat Sheet

Comprehensive Automation Guide for Java Developers

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1. TestNG Setup (Maven Dependency)

```
<dependency>
  <groupId>org.testng</groupId>
  <artifactId>testng</artifactId>
  <version>{7.9.0}</version>
  <scope>test</scope>
</dependency>
```

2. Basic TestNG Test

```
import org.testng.annotations.Test;

public class LoginTest {
    @Test
    public void loginTest() {
        System.out.println("Login Test");
    }
}
```

3. TestNG Annotations (MOST IMPORTANT)

Execution Flow:

@BeforeSuite → @BeforeTest → @BeforeClass → @BeforeMethod → @Test → @AfterMethod ...

```
@BeforeMethod
public void setup() { driver = new ChromeDriver(); }

@AfterMethod
public void tearDown() { driver.quit(); }
```

4. Priority, Groups and Enabled

```
@Test(priority = 1)
@Test(priority = 2, enabled = false)
@Test(groups = {"smoke", "regression"})
```

5. Depends On Method

```
@Test
public void login() {}

@Test(dependsOnMethods = "login")
public void dashboard() {}
```

6. Invocation Count (Repeat Test)

```
@Test(invocationCount = 3)
public void retryTest() {} // Runs 3 times
```

7. Timeout

```
@Test(timeOut = 5000)
public void timeLimitTest() {}
```

8. Assertions

```
Assert.assertEquals(actual, expected);
Assert.assertTrue(condition);
Assert.assertFalse(condition);
Assert.assertNotNull(object);
Assert.fail();
```

9. Hard vs Soft Assert

Hard Assert: `Assert.assertEquals(title, "Home");`

Soft Assert:

```
SoftAssert sa = new SoftAssert();
sa.assertEquals(title, "Home");
sa.assertTrue(isDisplayed());
sa.assertAll(); // Critical to report failures
```

10. Parameterization (testng.xml)

XML: `<parameter name="browser" value="chrome"/>`

Java:

```
@Parameters("browser")
@Test
public void runTest(String browser) { System.out.println(browser); }
```

11. DataProvider (MOST ASKED)

```
@DataProvider(name = "loginData")
public Object [][] getData() {
    return new Object [][] { {"user1", "pass1"}, {"user2", "pass2"} };
}

@Test(dataProvider = "loginData")
public void loginTest(String user, String pass) {}
```

12. Parallel Execution

```
<suite name="ParallelSuite" parallel="tests" thread-count="2">
```

13. Listeners (IMPORTANT)

```
public class TestListener implements ITestListener {
    public void onTestFailure(ITestResult result) {
        System.out.println("Test Failed: " + result.getName());
    }
}

// Attach Listener:
@Listeners(TestListener.class)
public class LoginTest {}
```

14. Retry Failed Test

```
public class RetryAnalyzer implements IRetryAnalyzer {
    int count = 0;
    public boolean retry(ITestResult result) {
        return count++ < 2;
    }
}

@Test(retryAnalyzer = RetryAnalyzer.class)
```

15. TestNG Reports

Found in test-output folder:

- [emailable-report.html](#)
- [index.html](#)

16. TestNG Interview Questions

- ✓ **@BeforeTest vs @BeforeClass**
- ✓ **DataProvider vs @Parameters**
- ✓ **Hard vs Soft Assert**
- ✓ **Listeners use case**
- ✓ **Parallel execution issues**

17. Selenium + TestNG Framework Structure

```
src/test/java
  base
    BaseTest.java
  pages
    LoginPage.java
  tests
    LoginTest.java
  utils
    ConfigReader.java
  listeners
    TestListener.java
```